Girls Build Bots — Junior Digital Game Design

Badge-aligned workshop guide for robotics-team-led outreach

Level: Juniors (Grades 4–5) · **Badge:** Coding for Good — Digital Game Design **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 90–120 minute session · **Ratio:** 1 mentor per 4–5 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools the team or troop can provide.

> **Troop Leaders:** This curriculum is designed to *align with* the Junior Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

Games are built *for players*. A good designer thinks about who plays, whether it's fair, and whether it's fun for everyone — positive gaming is the "for good" of this badge.

Vocabulary: mechanic, rule, goal, win condition, difficulty, playtest, iterate, positive gaming.

Digital Game Design (90–120 minutes)

Badge theme: Designing a playable digital game — mechanics, rules, goals — and improving it through playtesting.

Materials

- Laptops with **MakeCode Arcade** (arcade.makecode.com)
- Game design doc (one page): Goal / Rules / Win condition / What makes it fun / Who it's for
- Projector for live coding; sticky notes for playtest feedback

Activity 1 — Dissect a Game Like a Designer (15 min)

Pick a game everyone knows. Break it into designer parts: the **goal**, the **rules**, the **win condition**, and the **difficulty curve** (does it get harder as you go?). Add one more question designers ask: is it *fair and fun for every player*? That's positive gaming.

Activity 2 — Write the Design Doc (15 min)

Every real studio starts with a design document. Each Scout fills a one-pager for her game: goal, rules, win condition, what makes it fun, and who it's for. Mentors enforce scope: one goal, one enemy or obstacle, one way to win.

Activity 3 — Build the Game (55 min)

Progressive build in MakeCode Arcade: 1. Player sprite with arrow-key movement 2. Collectible with a score counter 3. **Win condition** — reach the target score, show "YOU WIN" 4. **Challenge element** — an enemy that chases, or a countdown timer 5. Stretch: a second level or increasing speed (the difficulty curve!)

Scouts implement *their* design doc, not a fixed clone — mentors help translate each Scout's idea into the same core blocks.

Activity 4 — Structured Playtest (15 min) + Wrap (10 min)

Swap seats. Testers play for two minutes, then leave two sticky notes: one ⭐ (what's fun) and one 💡 (one change that would make it better). Designers make **one improvement** based on feedback — that's iteration, the most important habit in game design. Closing: who changed their game after feedback? (Everyone should.)

> **Robot moment (optional):** Share for 60 seconds how a brand-new competition game drops every season and the team's first job is exactly this analysis — goals, rules, win conditions, strategy.

Completion Checklist — Digital Game Design

By the end of the session, to earn the badge, each Scout can: - [ ] Break a game into its designer parts: goal, rules, win condition, difficulty - [ ] Write a one-page design doc before building - [ ] Build a playable game with a win condition and a challenge element - [ ] Run a playtest and improve her game based on feedback

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Mentor Guide (for the robotics team)

- **You don't need a robot for this.** The "robot moment" box is a 60-second add-on.
- **Design doc first, always.** No screens until the one-pager is done — it's the difference between designing and doodling.
- **Protect scope ruthlessly but kindly.** Juniors will pitch five-level epics; help them ship one great level instead.
- **Feedback is a gift — teach the format.** One star, one wish, always both. Model receiving feedback gracefully yourself.
- **Name positive gaming.** Fun for *every* player, fair rules, no meanness — say it out loud when praising designs.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Junior Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.